Testing Linear Hypotheses

Lecture 4 (Stata code)

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Reading

This lecture covers material from Wooldridge (2025):

- Chapter 4.4 - "Testing Hypotheses about a Single Linear Combination of the Parameters"
- Chapter 4.5 - "Testing Multiple Linear Restrictions"
- Chapter 7.4 - "Interactions Involving Dummy Variables"

Note: I have edited the syllabus slightly to move the topic of heteroskedasticity to Week 5 with asymptotics.

Hypothesis Tests

Simple hypotheses

So far, we have considered hypotheses that involve a single estimator:

1. $H_0 : \beta_j = \kappa_0$
2. $H_0 : \beta_j \geq \kappa_0$
3. $H_0 : \beta_j \leq \kappa_0$

Linear combinations

Today, we consider hypotheses of the form:

$$H_0 : a_0\beta_0 + a_1\beta_1 + \cdots + a_k\beta_k = \kappa_0$$

for constants $\{a_j\}_{j=0,1,\ldots,k}$. This is a **linear combination of parameters**.

For example, $H_0 : \beta_1 = 2\beta_2 - 0.5$ can be written as,

$$H_0 : \underbrace{0}_{a_0}\,\beta_0 + \underbrace{1}_{a_1}\,\beta_1 + \underbrace{(-2)}_{a_2}\,\beta_2 = \underbrace{-0.5}_{\kappa_0}$$

Multiple linear combinations

Similarly, we could have q of these statements,

$$\begin{aligned} H_0 : a_0^1\beta_0 + a_1^1\beta_1 + \cdots + a_k^1\beta_k &= \kappa_0^1 \\ a_0^2\beta_0 + a_1^2\beta_1 + \cdots + a_k^2\beta_k &= \kappa_0^2 \\ &\vdots \\ a_0^q\beta_0 + a_1^q\beta_1 + \cdots + a_k^q\beta_k &= \kappa_0^q \end{aligned}$$

Single linear combinations

Consider the following example related to environmental and health economics

$$\ln(PM2.5_\ell) = \beta_0 + \beta_1 EV_\ell + \beta_2 hybrid_\ell + \mathbf{x}_\ell\gamma + u_\ell$$

where,

- $PM2.5$ is micrograms of particulate matter (with diameter ≤ 2.5 micrometres) per cubic metre in location ℓ.
  - standard measure for health effects as absorbable by lungs and bloodstream
- EV and $hybrid$ are number of registered vehicles (/1000) by fuel type in location ℓ.

Example: PM2.5

The following estimates use local authority level data for 2019 in the UK.

```
* Open CSV data
import delimited "$data_dir\pm25.csv", clear

* Estimate univariate linear regression
reg ln_pm25 ev_k hybrid_k petrol_k diesel_k ln_pop
```

```
(encoding automatically selected: ISO-8859-1)
(17 vars, 291 obs)

      Source |       SS           df       MS      Number of obs   =       291
-------------+----------------------------------   F(5, 285)       =     33.45
       Model |  2.84542086         5  .569084172   Prob > F        =    0.0000
    Residual |  4.84835963       285  .017011788   R-squared       =    0.3698
-------------+----------------------------------   Adj R-squared   =    0.3588
       Total |  7.69378049       290  .026530278   Root MSE        =    .13043

------------------------------------------------------------------------------
     ln_pm25 | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
        ev_k |   .0779783     .03734     2.09   0.038     .0044811    .1514755
    hybrid_k |   .0322276   .0063524     5.07   0.000      .019724    .0447312
    petrol_k |   .0000559   .0007053     0.08   0.937    -.0013323    .0014442
    diesel_k |   -.003546   .0005762    -6.15   0.000    -.0046802   -.0024119
      ln_pop |    .088135   .0208428     4.23   0.000     .0471096    .1291604
       _cons |   1.264753   .2325884     5.44   0.000     .8069436    1.722562
------------------------------------------------------------------------------
```

Null hypotheses

From a public policy perspective (if you consider designing subsidies), you want to estimate both β_1 and β_2, BUT also know if

$$H_0 : \beta_1 = \beta_2$$

That is, should the policy discriminate bewteen electrified fuel types?

- Engineers may be able provide mechanical differences, but they can't tell you about differences in usage (driver behaviour) which is important for realized pollution mitigation.

> **Warning**
>
> Do not take these estimates too seriously. This model is a classic example of selection into "treatment": those who live in higher pollution areas (e.g. dense urban centres) are also more likely to buy EVs. A better research design would use panel data to evaluate this question dyanmically.

Alternative hypothesis

The alternative in this case, may be

$$H_1 : \beta_1 \neq \beta_2$$

But may also be,

$$H_1 : \beta_1 < \beta_2$$

or

$$H_1 : \beta_1 > \beta_2$$

if there is a strong case for ruling out one case (see Wooldridge 2025, 139).

Test statistic

For $H_0 : \beta_1 = \beta_2 \Rightarrow H_0 : \beta_1 - \beta_2 = 0$

$$\text{test-stat} = \frac{\hat{\beta}_1 - \hat{\beta}_2}{\sqrt{Var(\hat{\beta}_1 - \hat{\beta}_2)}}$$

> **Note**
>
> The more general case would be $H_0 : a_1\beta_1 + a_2\beta_2 = \kappa_0$

$$\text{test-stat} = \frac{a_1\hat{\beta}_1 + a_2\hat{\beta}_2 - \kappa_0}{\sqrt{Var(a_1\hat{\beta}_1 + a_2\hat{\beta}_2)}}$$

which can be extended to include more parameters.

Variance

The key to understanding and executing this test is,[1]

$$Var(\hat{\beta}_1 - \hat{\beta}_2) \neq Var(\hat{\beta}_1) + Var(\hat{\beta}_2)$$

This is because,

$$\begin{aligned} Var(a_1z_1 + a_2z_2) =& Var(a_1z_1) + Var(a_2z_2) + 2 \times Cov(a_1z_1, a_2z_2) \\ =& a_1^2 Var(z_1) + a_2^2 Var(z_2) + 2 \times a_1a_2 Cov(z_1, z_2) \end{aligned}$$

for random variables z_1 and z_2 and constants a_1 and a_2.

So,

$$Var(\hat{\beta}_1 - \hat{\beta}_2) = Var(\hat{\beta}_1) + Var(\hat{\beta}_2) - 2\times \underbrace{Cov(\hat{\beta}_1, \hat{\beta}_2)}_{\neq 0}$$

Why -2?

- For this H_0,

$$\Rightarrow H_0 : \underbrace{\beta_1 - \beta_2 = 0}_{a_1\beta_1 + a_2\beta_2 = \kappa_0}$$

  So, $a_1 = 1$ and $a_2 = -1$.

[1] We drop the $Var(\cdot|X)$ notation from Lecture 3. However, all these variances are conditional on the sample values of the independent variables.

Distribution

Under MLR 1-6, the **conditional** distribution of this test-statistic is t_{n-k-1}

$$t\text{-stat} = \frac{\hat{\beta}_1 - \hat{\beta}_2}{\sqrt{Var(\hat{\beta}_1 - \hat{\beta}_2)}} \underset{H_0}{\sim} t_{n-k-1}$$

under the null.

- You can use the same steps from Lecture 3 to conduct inference: p-value, critical value.
- We tend not to construct confidence intervals in such cases.
- Similarly, you can compute the power under the alternative $\beta_1 - \beta_2 = \kappa$.

Example: PM2.5

```
* t-test
lincom ev_k - hybrid_k
```

```
 ( 1)  ev_k - hybrid_k = 0

------------------------------------------------------------------------------
     ln_pm25 | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         (1) |   .0457507   .0423487     1.08   0.281    -.0376053    .1291066
------------------------------------------------------------------------------
```

Multiple linear hypotheses

Another common form of hypothesis concerns **multiple** linear combinations of estimators.

$$\begin{aligned} H_0 : a_0^1\beta_0 + a_1^1\beta_1 + \cdots + a_k^1\beta_k &= \kappa_0^1 \\ a_0^2\beta_0 + a_1^2\beta_1 + \cdots + a_k^2\beta_k &= \kappa_0^2 \\ &\vdots \\ a_0^q\beta_0 + a_1^q\beta_1 + \cdots + a_k^q\beta_k &= \kappa_0^q \end{aligned}$$

Linear transformation

You can think of this as a linear transformation of β.

$$H_0 : R\beta = r$$

Where R is a $q \times k$ matrix of constants and r a q-dimensional column vector of constants.

Example: PM2.5

In our example of

$$PM2.5_\ell = \beta_0 + \beta_1 EV_\ell + \beta_2 hybrid_\ell + \mathbf{x}_\ell\gamma + u_\ell$$

This could be as simple as,

$$H_0 : \beta_1 = 0 \text{ and } \beta_2 = 0$$

Multiple hypothesis testing

Why not test $H_0^{(1)} : \beta_1 = 0$ and then $H_0^{(2)} : \beta_2 = 0$ as separate individual hypotheses? The combined size of these tests would not be α.

$$Pr(\text{Reject } H_0^{(1)} \text{ or } H_0^{(2)} | \beta_1 = 0, \beta_2 = 0) \neq \alpha$$

Why? For two reasons:

- Suppose, the tests are independent. Then,

$$\begin{aligned} Pr(\text{Reject } H_0^{(1)} \text{ or } H_0^{(2)} | \beta_1 = 0, \beta_2 = 0) =& Pr(\text{Reject } H_0^{(1)} | \beta_1 = 0, \beta_2 = 0) \\ &+ Pr(\text{Reject } H_0^{(2)} | \beta_1 = 0, \beta_2 = 0) \\ =& 2\alpha \end{aligned}$$

  We could then adjust the size of each individual test to $\alpha/2$ to ensure that the probability of making *at least one* type-1 error is α overall.

- The bigger ("messier") problem is that the tests are not independent of one another because the $Cov(\hat{\beta}_1, \hat{\beta}_2) \neq 0$. This makes it difficult to define this probability.

Multiple hyopthesis testing

There are a number of ways to correctly do multiple hypothesis testing. One is the Bonferroni correction which gets you to reject separate hypotheses sequentially using an algorithim. It does so by ensuring that the family-wise error rate (the probability of rejecting at least one true H_0 across a "family" of hypotheses) is $\leq \alpha$.

Test

The test is performed by considering two models:

1. **Unrestricted** model; using our example,

$$\ln(PM2.5_\ell) = \beta_0 + \beta_1 EV_\ell + \beta_2 hybrid_\ell + \mathbf{x}_\ell\gamma + u_\ell$$

- storing the SSR_{ur} from this model

2. **Restricted model**, which imposes $H_0 : \beta_1 = 0$ and $\beta_2 = 0$,

$$\ln(PM2.5_\ell) = \beta_0 + \mathbf{x}_\ell\gamma + u_\ell$$

- storing the SSR_r from this model

Test statistic

The test static that is then given by,

$$F\text{-stat} = \frac{(SSR_r - SSR_{ur})/q}{SSR_{ur}/df_{ur}}$$

where,

- SSR_r and SSR_{ur} come from the restricted (r) and unrestricted models (ur).[2]

$$SSR_r - SSR_{ur} > 0$$

[2]This inequality is how you can remember the order. The F-stat is strictly positive.

Test statistic

The test static that is then given by,

$$F\text{-stat} = \frac{(SSR_r - SSR_{ur})/q}{SSR_{ur}/df_{ur}}$$

and,

- q is the number of restrictions (i.e. equations tested) and is given by the difference in residual degrees of freedom

$$q = df_r - df_{ur}$$

Distribution

Under the null, F-stat has an F distribution (hence the name)

$$F\text{-stat} \underset{H_0}{\sim} F_{q,n-k-1}$$

We reject H_0 if

$$F\text{-stat} > F^{-1}_{q,n-k-1}(1-\alpha) = f_{q,n-k-1,1-\alpha}$$

where $F^{-1}_{q,n-k-1}(1-\alpha)$ is the $1-\alpha$ percentile of the distribution.

Rejection rule

$$F\text{-stat} > f_{q,n-k-1,1-\alpha}$$

- **Always a one-sided test**.[3]
- Doesn't mean that H_1 is one-sided.
  - You cannot tell which direction the deviation from the null is, because F-stat uses squared-deviations (i.e. sign is lossed) .
- $H_1:$ H_0 false, or $H_1 : \beta_1 \neq 0$ or $\beta_2 \neq 0$

[3]This does not mean that F-distributed tests are always one-sided, but rather that multiple linear hypotheses of this form are.

F distribution

The F distribution is defined as the ratio of two chi-squared distributions,

$$F_{q,m} = \frac{\chi^2_q/q}{\chi^2_m/m}$$

each normalized by their corresponding degrees of freedom.

Alternative form

Since $SSR = (1 - \mathbf{R}^2)SST$ and $SST_r = SST_{ur}$,

We can write the F-stat as,

$$F\text{-stat} = \frac{(\mathbf{R}^2_{ur} - \mathbf{R}^2_r)/q}{(1 - \mathbf{R}^2_{ur})/(n - k - 1)}$$

Note,

- The order switches in the numerator, but remains positive since $\mathbf{R}^2_{ur} > \mathbf{R}^2_r$
- $n - k - 1 = df_{ur}$, with k the number of parameters in the unrestricted model

Example: PM2.5

```
* F-test
test ev_k hybrid_k
```

```
 ( 1)  ev_k = 0
 ( 2)  hybrid_k = 0

       F(  2,   285) =   53.88
            Prob > F =    0.0000
```

Revisiting single linear restriction

You can think of a single linear hypothesis as a special case of multiple where $q = 1$. For example,

$$H_0 : \beta_1 = \beta_2$$

- Restricted model:

$$\begin{aligned}\ln(PM2.5_\ell) =& \beta_0 + \beta_1 EV_\ell + \underbrace{\beta_1}_{=\beta_2} hybrid_\ell + \mathbf{x}_\ell\gamma + u_\ell \\ =& \beta_0 + \beta_1(EV_\ell + hybrid_\ell) + \mathbf{x}_\ell\gamma + u_\ell\end{aligned}$$

  A regression with a new transformed variable: $EV_\ell + hybrid_\ell$

Moreover, you can show that for $q = 1$

$$F\text{-stat} = \frac{SSR_r - SSR_{ur}}{SSR_{ur}/(n-k-1)} = (t\text{-stat})^2$$

Example: PM2.5

```
* t-test
lincom ev_k - hybrid_k

* F-test
test ev_k=hybrid_k

* check
dis sqrt(r(F))
```

```
 ( 1)  ev_k - hybrid_k = 0

------------------------------------------------------------------------------
     ln_pm25 | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         (1) |   .0457507   .0423487     1.08   0.281    -.0376053    .1291066
```

```
------------------------------------------------------------------------------

 ( 1)  ev_k - hybrid_k = 0

       F(  1,   285) =    1.17
            Prob > F =    0.2809

1.0803322
```

Regression significance

A special case of this form of test concerns the joint significance of all the *slope* parameters in a model:

$$H_0 : \beta_1 = 0,\ \beta_2 = 0,\ \ldots,\ \beta_k = 0$$

In this case,

- The restricted model is a regression on a constant (alone) where $\Rightarrow \mathbf{R}^2_r = 0$
- And $\mathbf{R}^2_{ur}$ is simply the $\mathbf{R}^2$ from the model
- The number of restrictions is $q = k$

We define the F-stat as,

$$F\text{-stat} = \frac{\mathbf{R}^2/k}{(1-\mathbf{R}^2)/(n-k-1)}$$

Stata output

This test is reported as standard with the Stata output as:

- `F( , )`: F-stat
- `Prob > F`: corresponding p-value of F-test

```
      Source |       SS           df       MS      Number of obs   =       291
-------------+----------------------------------   F(5, 285)       =     33.45
       Model |  2.84542086         5  .569084172   Prob > F        =    0.0000
    Residual |  4.84835963       285  .017011788   R-squared       =    0.3698
-------------+----------------------------------   Adj R-squared   =    0.3588
       Total |  7.69378049       290  .026530278   Root MSE        =    .13043

------------------------------------------------------------------------------
     ln_pm25 | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
        ev_k |   .0779783     .03734     2.09   0.038     .0044811    .1514755
    hybrid_k |   .0322276   .0063524     5.07   0.000      .019724    .0447312
    petrol_k |   .0000559   .0007053     0.08   0.937    -.0013323    .0014442
    diesel_k |   -.003546   .0005762    -6.15   0.000    -.0046802   -.0024119
      ln_pop |    .088135   .0208428     4.23   0.000     .0471096    .1291604
       _cons |   1.264753   .2325884     5.44   0.000     .8069436    1.722562
------------------------------------------------------------------------------
```

Structural Change

Dummy variables: recap

Recall, when we have the model:

$$y_i = \beta_0 + \beta_1 d_i + u_i$$

where $d_i \in \{0, 1\}$. Then,

$$\beta_1 = E[y_i|d_i = 1] - E[y_i|d_i = 0]$$

Dummy variable interactions

Now, suppose the model was

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 d_i + \beta_3 x_i \times d_i + u_i$$

where x_i is continuous.

- What is the interpretation of each parameter?

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 d_i + \beta_3 x_i \times d_i + u_i$$

- $\beta_0 = E[y_i | d_i = 0, x_i = 0]$
- $\beta_1 = \frac{dE[y_i | d_i = 0, x_i]}{dx_i}$
- $\beta_2 = E[y_i | d_i = 1, x_i = 0] - E[y_i | d_i = 0, x_i = 0]$
- $\beta_3 = \frac{dE[y_i | d_i = 1, x_i]}{dx_i} - \frac{dE[y_i | d_i = 0, x_i]}{dx_i}$

Another way of thinking about it:

- For $d_i = 0$, the model is:

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

- For $d_i = 1$, the model is:

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 + \beta_3 x_i + u_i$$

  Which can be rewritten as:

$$y_i = \underbrace{(\beta_0 + \beta_2)}_{\gamma_0} + \underbrace{(\beta_1 + \beta_3)}_{\gamma_1} x_i + u_i$$

We have two models: for $d_i = 0$ and $d_i = 1$.

- β_2 is the difference in intercept
- β_3 is the difference in slope

Example

The typical example is the returns to education for men and women (as in Wooldridge 2025, 243)

- But, given what we know about the 'parent penalty' (Kleven et al. 2019), it seems more pertinent to focus on experience

$$\ln(wage_i) = \beta_0 + \beta_1 exp_i + u_i$$

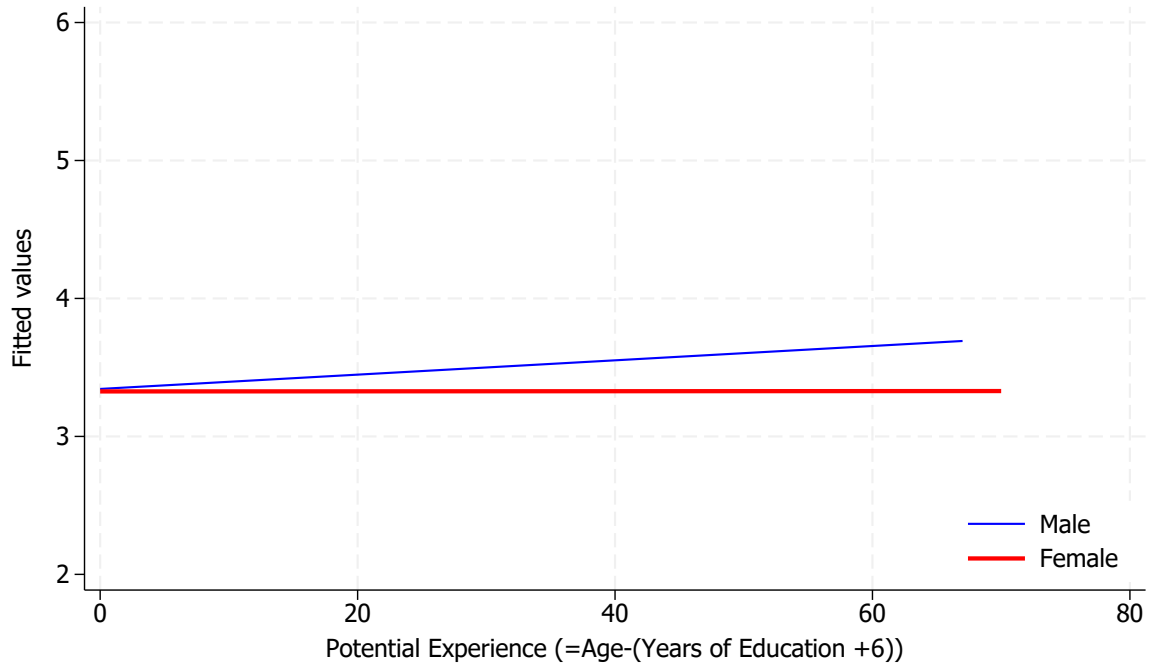
```
use "$data_dir\morg24_bothgen.dta", clear

* Scatter
twoway (scatter lnwage exp, mc(blue%50)) (lfit lnwage exp, lwidth(*2)), by(sex,  legend(off)
```



If we overlay the fitted lines we get

```
* overlay fitted lines
twoway (lfit lnwage exp if sex==1, lc(blue)) (lfit lnwage exp if sex==2, lc(red) lwidth(*2))
```



F-test

The *F*-test can be used to test differences in a model across groups.[4]

- For example, whether the Mincer equation (from Tutorial 1) differs by gender.

$$\ln(wage_i) = \beta_0 + \beta_1 edu_i + \beta_2 exp_i + u_i$$

- We estimated this model for the male-only sample
- The quadratic term in *exp* has been dropped for simplification

[4]We will look at the case of two groups, but the approach can be extended to multiple groups.

- The model has $k + 1 = 3$ parameters

Test:

We may wish to ask whether:

- the model for men (group = 1; e.g. "Male")

$$\ln(wage_i) = \beta_0 + \beta_1 edu_i + \beta_2 exp_i + u_i$$

is the same as,

- the model for women (group = 2; e.g. "Female")?

$$\ln(wage_i) = \gamma_0 + \gamma_1 edu_i + \gamma_2 exp_i + u_i$$

Null hypothesis

This question implies a null hypothesis:

$$H_0 : \beta_0 = \gamma_0,\ \beta_1 = \gamma_1,\ \beta_2 = \gamma_2$$

against,

$$H_1 : H_0 \text{ false}$$

Chow test

The relevant F-statistic for this test is referred to as the Chow statistic:

$$F\text{-stat} = \frac{(SSR_P - (SSR_1 + SSR_2))/(k+1)}{(SSR_1 + SSR_2)/(n - 2(k+1))}$$

where,

- SSR_P is the "pooled" (P) SSR from a regression of the joint sample.
- SSR_1 is the SSR from a regression for just group 1.
- SSR_2 is the SSR from a regression for just group 2.

  - together SSR_1 and SSR_2 represent the unrestricted model.
- $n - 2(k+1)$ is the **combined** degrees of freedom from group 1 and 2.
  - $k+1$ parameters are estimated in each group.

Distribution

The F-stat is (unsurprisingly) F-distributed,

$$F\text{-stat} \underset{H_0}{\sim} F_{k+1,n-2(k+1)}$$

Approach 2

This test has an equivalent form using interactions. Let,

$$d_i = \mathbf{1}\{\text{'Group 2'}\}$$

For example,

$$d_i = \mathbf{1}\{\text{'Female'}\}$$

Interactions

We can write the unrestricted model as,

$$\ln(wage_i) = \beta_0 + \beta_1 edu_i + \beta_2 exp_i + \beta_3 d_i + \beta_4 edu_i \times d_i + \beta_5 exp_i \times d_i + u_i$$

We now have 6 parameters: $2 \times (k+1)$ (for $k = 2$)

As before, the dummy and dummy-interactions tell you about differences,

$$\beta_3 = E[\ln(wage_i)|d_i = 1, edu_i = 0, exp_i = 0] - E[\ln(wage_i)|d_i = 0, edu_i = 0, exp_i = 0]$$

and

$$\beta_j = \frac{\partial E[\ln(wage_i)|d_i = 1, edu_i, exp_i]}{\partial x_{ij}} - \frac{\partial E[\ln(wage_i)|d_i = 0, edu_i, exp_i]}{\partial x_{ij}} \qquad j = 4, 5$$

Test

Intuitively, the test for whether the model differs by group is given by,

$$H_0 : \beta_3 = 0,\ \beta_4 = 0,\ \beta_5 = 0$$

where,

- **unrestricted** model is given by,

$$\ln(wage_i) = \beta_0 + \beta_1 edu_i + \beta_2 exp_i + \beta_3 d_i + \beta_4 edu_i \times d_i + \beta_5 exp_i \times d_i + u_i$$

- $df_{ur} = n - 6$

- **restricted** model is given by,

$$\ln(wage_i) = \beta_0 + \beta_1 edu_i + \beta_2 exp_i + v_i$$

- $df_r = n - 3$

F-stat

The F-stat is given by,

$$F\text{-stat} = \frac{(SSR_r - SSR_{ur})/3}{SSR_{ur}/(n-6)}$$

Compare this to the first approach:

$$F\text{-stat} = \frac{(SSR_P - (SSR_1 + SSR_2))/3}{(SSR_1 + SSR_2)/(n-6)}$$

Equivalence

These two test-statics are equivalent because,

1. $SSR_P = SSR_r$
2. $SSR_1 + SSR_2 = SSR_{ur}$

Demonstration

Here are the SSR for each version:

- Model for group 1 (i.e. "male")

```
qui reg lnwage edu exp if sex==1
dis e(rss)
```

```
715.90864
```

- Model for group 2 (i.e. "female")

```
qui reg lnwage edu exp if sex==2
dis e(rss)
```

```
539.59768
```

- Unrestricted model (i.e. with interactions)

```
qui reg lnwage i.sex##c.edu i.sex##c.exp
dis e(rss)
```

```
1255.5063
```

- Restricted model (i.e. without interactions)

```
qui reg lnwage edu exp
dis e(rss)
```

```
1307.1667
```

Here is how you would programme the test

- Estimate interacted model

```
qui reg lnwage i.sex##c.edu i.sex##c.exp
test 2.sex 2.sex#c.edu 2.sex#c.exp
```

```
 ( 1)  2.sex = 0
 ( 2)  2.sex#c.edu = 0
 ( 3)  2.sex#c.exp = 0

       F(  3,  4994) =   68.50
            Prob > F =    0.0000
```

The notation `2.sex` is how Stata names the coefficient on level `2` of the factor variable `sex`, and `2.sex#c.edu` is the interaction with `c.edu` (a continuous variable).

References

Bibliography

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